

Immersive Reading: Comparison of Performance and User Experience for Reading Long Texts in Virtual Reality

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ABSTRACT

Specific use cases in virtual reality (VR) require users to read long texts on their headsets. We use VR to support researchers in the humanities, which includes the display of long text descriptions in VR. However, research on suitable user interface (UI) patterns for displaying and interacting with long texts in VR is scarce. To address this gap, we designed four text panel variants and conducted a within-participants study (N=24) to evaluate user experience (UX) and reading performance. Our findings suggest that there are no significant differences between conditions regarding reading performance. Yet, there are significant effects of conditions on some aspects of UX. Our work offers initial insights and future directions for research on the design of suitable UI patterns and the UX for reading long texts in VR.

CCS CONCEPTS

• **Human-centered computing** → **Virtual reality**; *Graphical user interfaces*; Interactive systems and tools.

KEYWORDS

user-centred design, virtual reality, reading, interface design patterns

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1 INTRODUCTION

With the increasing improvement of virtual reality (VR) technologies, novel applications scenarios arise [19]. For instance, due to the improved head-mounted display (HMD) quality and resolution, it is possible to read even large amounts of textual information in VR. In the scope of our project at the Cluster of Excellence “Understanding Written Artefacts” at the Universität Hamburg, we develop VR applications for humanity researchers. These researchers study

numerous ancient inscriptions, so-called “Written Artefacts”, which are annotated with long textual paragraphs [6, 7]. However, there is still limited research on reading longer text in VR, especially regarding user experience (UX) and suitable user interface (UI) patterns and guidelines.

In our work, we contribute research on reading performance and UX for reading long texts in VR. To investigate possible suitable UI configurations, we developed four variants of a text panel for displaying and interacting with long texts in VR in a user-centred design approach. The text panels were created based on 2D desktop UI patterns, which are already familiar to our target user group. Two of the panel variants use a continuous UI pattern and the other two use a discrete one: the *continuous variants* display the text within their view port in one continuous flow that users can scroll through via a scrollbar or buttons. The text in the *discrete variants* can be navigated through page by page according to the pagination pattern. Due to their differences in spatial layout and interaction requirements, it is reasonable to assume that these panels affect reading performance as well as the UX. Research on reading on desktop computers showed no difference in using scrolled versus paginated text [1], but, to our best knowledge, this has not yet been studied for reading in VR. Therefore, we evaluated *Recall*, *Spatial Recall*, and UX for the four text panel variants in a within-participants design to investigate potential differences between discrete and continuous text representations in VR. We tested the following three hypotheses:

- H1 Discrete text display has a significant better *Recall* than continuous text display for reading long texts in VR.
- H2 Discrete text display has a significant better *Spatial Recall* than continuous text display for reading long texts in VR.
- H3 Discrete text display has a significant better UX than continuous text display for reading long texts in VR.

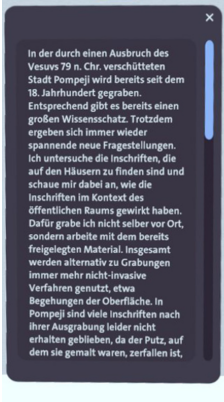
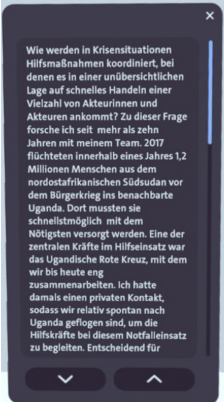
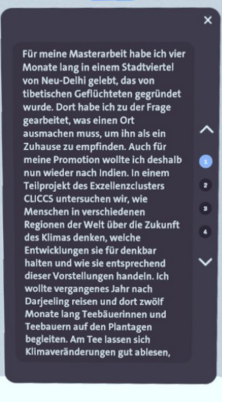
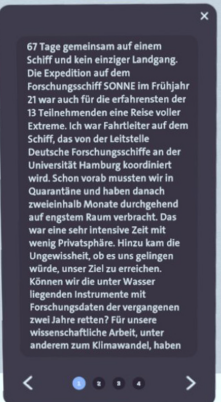
Participants had to read four different but equivalent texts on each of the text panels variants, answer three content questions, and indicate the position of the information in the text. We also collected subjective rankings and qualitative user feedback with open-ended questions.

2 RELATED WORK

Previous work indicates that reading performance in VR might differ from that of a conventional 2D display [17]. For instance, studies comparing reading on printed paper versus reading on screens show that the former resulted in significantly better reading comprehension, accuracy, and speed [2, 13]. Moreover, Wästlund et al. [24] reported a higher cognitive load when reading and working on the

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Table 1: Four text panel variants with different UI controls using modified text excerpts from our university magazine. [22]

CONTINUOUS Text Representation		DISCRETE Text Representation	
C.BAR	C.BUT	D.VER	D.HOR
			

computer within the same classroom. Rau et al. [17] found that participants took about 10% longer when answering multiple-choice questions after reading on VR and augmented reality (AR) HMDs compared to reading on conventional desktop LCD screens. In their later work, Rau et al. [16] found that reading in VR and AR HMDs was slower than on an LCD monitor. Their participants spent significantly more time reading in VR and AR HMDs to achieve similar accuracy to reading on LCD screens.

Kojić et al. [11] evaluated user experience for reading in VR and tested values for different text parameters. In their study, participants could adjust different parameters such as font size, distance, and contrast on two HMDs, the Oculus Quest and Go, and for three text samples: short, medium and long. Their findings revealed that there are significant differences for preferred values of angular size depending on the length of the text, while the two different device types had no significant influence on the text parameters. Kobayashi et al. [10] investigated text parameters for reading long texts in the Japanese language and give recommendations for font type, size, and reading distance. Moreover, all their participants preferred text panels that were placed at a fixed position in VR instead of panels that were following the user's position and view. Rzaev et al. [18] observed the same preference for fixed panels for reading long texts in VR. Moreover, they recommend displaying long texts in one paragraph. Several studies on reading in VR all reported that light text on a dark background was preferred by the participants [3, 4, 10].

3 TEXT PANEL DESIGNS

The requirements and design for the text panels were identified and developed with a user-centred design approach. We worked with two researchers from the humanities as end-users for our research focused VR application. Both were interviewed twice regarding their requirements and expectations for reading long texts in VR. We also tested the design iterations of the text panels with them.

The researchers had no prior experience with VR, apart from using the Meta Quest 2 as part of our interdisciplinary project. They usually work with analogue materials and 2D data on their computer and are not familiar with 3D applications. Accordingly, the UI controls for the text panels should be usable by humanities researchers without prior VR and 3D experience. We used simple UI elements that were familiar to the researchers, scrollbars and buttons, and tested panel variants with different UI controls: a vertical scrollbar, up and down buttons at the bottom of the panel for vertically scrolling the text, the same up/down buttons but for turning individual pages, and pagination controls with page numbers on buttons. The researchers liked the uninterrupted text flow of the scrolled text, the scrolling speed and controls of the scroll buttons, and the general overview and structure of the paginated text. Based on their feedback, we combined and improved the UI controls, and adapted two established 2D desktop UI patterns for our use case: continuous scrolling and discrete pagination. [20, 21]. Text panels using established desktop UI patterns can be easily understood by users without 3D and VR experience due to the familiar look and interaction style of the interface [8].

We created four final variants, consisting of two continuous and two discrete variants for the text representation and interaction, as seen in Table 1. In the *continuous variants* C.BAR (scrollbar) and C.BUT (scrollbutton) the text is visualized as a continuous flow with one part of the text displayed within a predefined viewport. Users can freely and continuously scroll up and down in the text. For easier interaction, the scrollbar's selectable bounds have an increased width compared to the displayed visuals (UI sprite). The size and position of the scrollbar indicates the total text amount and the current position. Additionally, up/down buttons can be used for scrolling in C.BUT. In the *discrete variants* D.VER (vertical) and D.HOR (horizontal) the text is divided into individual segments that can be navigated through page by page via the numbered pagination and arrow buttons. Both variants only differ in the positioning of their UI controls and the direction of the page turning animation



Figure 1: VR study environment and stationary study setup

(vertical vs. horizontal movement of the virtual pages).

The text panels are implemented on *World Space Canvases* in Unity 2020.3 LTS. We used light text (#FFFFFF) on dark background (panel #423E47, text area #3A343E) on all four designs based on findings saying that this leads to less fatigue and is preferred by readers over the reverse colour scheme [3, 4]. For text rendering and formatting, we used the TextMeshPro package included in Unity.

4 USER STUDY

For this preliminary study, our aim was to obtain initial, generally applicable findings for reading long texts in VR. We conducted an empirical user study in VR to evaluate the four text panel variants (see Table 1) regarding performance and UX. To test the hypotheses stated in Section 1, we examined whether there were differences between discrete and continuous text representations.

4.1 Text Parameters

To ensure that our results in the four conditions are comparable with each other, we chose four different non-scientific texts with a similar Flesch Reading Ease (FRE) [5]. The texts had the same overall topic – researchers presenting their international research projects. FRE measures the readability and comprehensibility of a text independently of its content. We used modified excerpts from an old issue of our university’s magazine [22] from 2021. The texts were in German, the native language of the participants. The FRE for German language is calculated according to Equation (1) [15]:

$$FRE_{German} = 180 - ASL - (58,5 * ASW) \quad (1)$$

where

ASL : *Average Sentence Length*, average no. of words per sentence

ASW : *Average Number of Syllables per Word*

The selected texts have a FRE between 47.4–50.8 ($M = 49.3$, $SD = 1.99$), which is rated as moderately difficult to read. All texts have 76 rows with a total word count of 351–366 ($M = 359.75$, $SD = 5.85$) and a *Mean* of 4.6–4.8 words per row. Each text is displayed in a single paragraph on a UI panel in portrait view, with 24 rows of text within its viewport. The UI panel’s position is fixed in *World Space* in VR according to the findings of Rzaev et al. [18] and Kobayashi et al. [10]. We used our university’s sans serif font with *Regular* font weight and positioned the information panels at a distance of about 1.75 m from the user. We chose a font size corresponding to

17.87 dmm based on the findings of Kojić et al. [11] for displaying long texts in the Quest head-mounted displays. The colour contrast ratio between texts to background is 11.87 : 1, which is well over the minimum contrast ratio of 4.5 : 1 defined by the World Wide Web Consortium (W3C) [23].

4.2 Procedure

We used a within-participant study design with four conditions corresponding to the text panel variants introduced in Section 3: *C.BAR*, *C.BUT*, *D.VER*, *D.HOR*. We randomized the order in which the conditions were shown to the participants, as well as the order of the four texts (Text 1–Text 4) that were assigned to the text panel variants. To counterbalance the condition order and the assignment of the text for each condition, we used Zeelenberg and Pecher’s [25] method for constructing Latin squares.

We tested 24 participants (12 female, 12 male) with 8 participants aged between 18–24, 12 between 25–34, and 4 between 35–44. The participants were recruited from our local university and were mainly students or researchers in the field of human-computer interaction. All had normal or corrected to normal vision and were native German speakers. 15 participants had used a VR HMD at least once, while 7 had no VR experience. The study took approximately 60 minutes in total and was conducted in a seated, stationary VR setup as pictured in Figure 1. In the beginning, participants signed a consent form and were informed about the procedure of the study. We used a Meta Quest 2 HMD and Quest Touch Controllers. Participants without VR experience received an introduction. Participants were instructed to read the texts carefully. They were informed that they had to answer multiple content questions and mark the location of the corresponding information in the text on printouts. These printouts had stylized text lines and line numbers. We handed out a sample to explain its usage. We also asked the participants to pay attention to the user interface of the text panels. Before entering the VR study environment, participants completed a demographic questionnaire and an initial Simulator Sickness Questionnaire (SSQ) [9] for measuring the baseline. After reading each text, we recorded the *Recall* of information as well as the *Spatial Recall* for the information’s position as a measure for reading performance. To account for serial positional effects, we asked questions from the beginning, middle, and the end of each text. To keep the difficulty level of the questions as similar as possible, we asked straightforward questions that could be answered

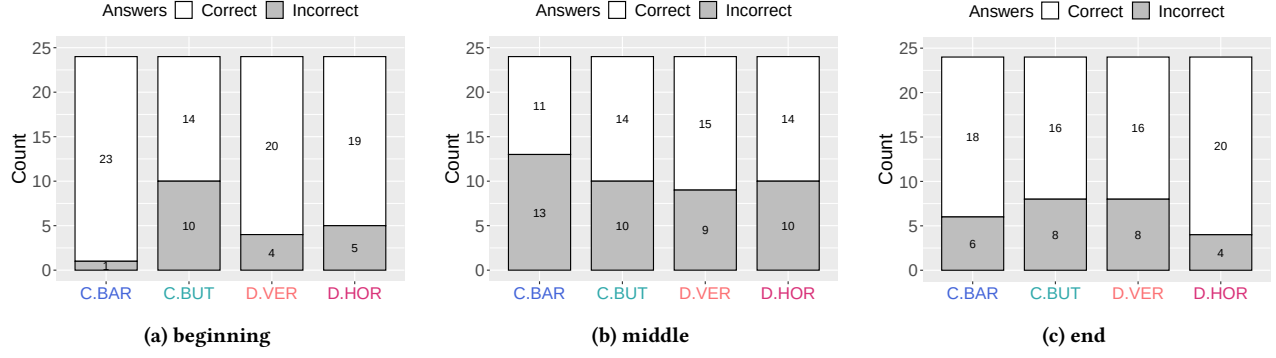


Figure 2: Information Recall: Correct and incorrect answers to the content questions for each panel variant. Plotted for the beginning, middle, and end of the texts.

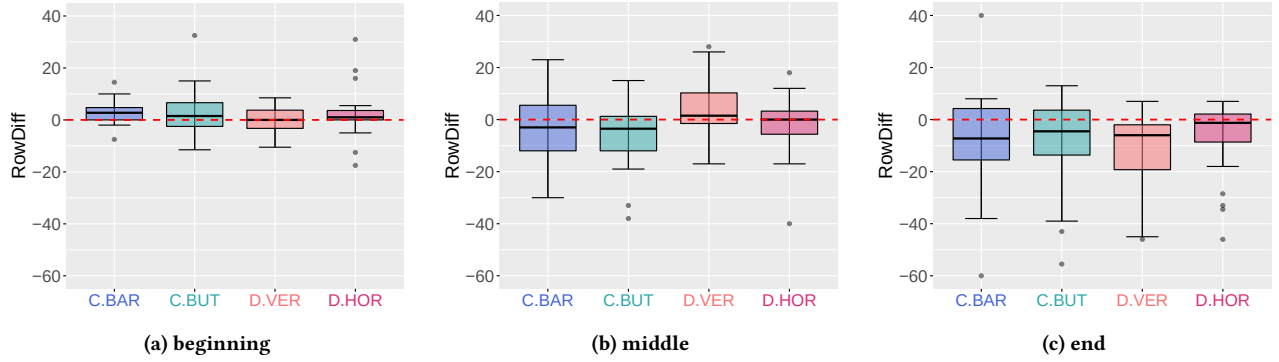


Figure 3: Spatial Recall: Row difference *RowDiff* between the correct row number containing the answer to each question and the number stated by the participants. Plotted for the beginning, middle, and end of the texts for each panel variant.

with a short answer. For example, we asked about places, e.g. a city or geographic location or about specific events, names, and equipment used. Unlike Rau et al. [16, 17] we used open-ended text fields for answering the content questions instead of multiple-choice questions to avoid giving the participants any clues. In addition, participants marked the corresponding spatial position of each answer on the printout. Participants also filled out the SSQ and the User Experience Questionnaire (UEQ) [12] after each trial. At the end of the study, we collected additional qualitative feedback. Participants gave a subjective ranking for the text panel variants. They were asked to explain their ranking and list positive and negative aspects of each variant. All questionnaires and the printouts were filled out on a PC at a separate desk. Participants were asked to recentre the view on the HMD each time after they put the Quest 2 back on.

5 RESULTS

We performed both quantitative and qualitative analyses of the data from the questionnaires and printouts. First, we present the qualitative results regarding *Recall* and *Spatial Recall*, followed by the results of the standardized questionnaires. Subsequently, we show the subjective user ranking for text panel variants and the qualitative analysis of our open-ended questions.

5.1 Recall

As seen in Figure 2, more content questions were correctly answered at the beginning and the end compared to the middle of the text for all text panel variants. The Cochran Q-test found no significant differences between the four variants for the middle and end of the text, but there was a statistically significant difference in *Recall* for the beginning ($p = .021$). According to the post-hoc McNemar test with Bonferroni correction, there is a significant difference between C.BAR and C.BUT ($p = .04$) with C.BAR having a ratio of correct to incorrect answers of 23 : 1 compared to C.BUT with 14 : 10.

5.2 Spatial Recall

We compared the row difference *RowDiff* between the correct row number of the answer to a content question and the participant's specified row number. So, a correct *Spatial Recall* would have *RowDiff* = 0 and a localization too late in the text would have a positive *RowDiff*, while negative *RowDiff* indicates a localization too early in the text. Figure 3 shows that *Spatial Recall* within the interquartile range (IQR) generally lies between +10 and -10 *RowDiff*, except at the end of the text, where *Spatial Recall* is heavily skewed towards the negative numbers.

Based on the Shapiro-Wilk test, a normal distribution could not be assumed for *RowDiff*. The Friedman test and consecutive post-hoc

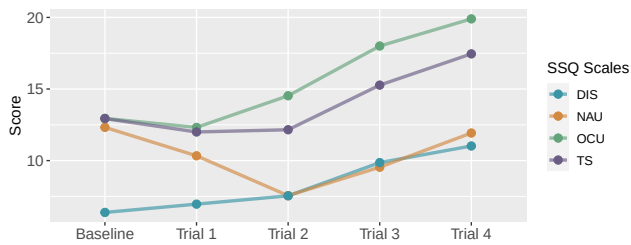
Table 2: Mean ratings of the six UEQ scales for each panel variant

	Attractiveness	Perspicuity	Efficiency	Dependability	Stimulation	Novelty
C.BAR	0.125	1.302	0.927	0.490	0.240	-0.969
C.BUT	0.958	1.510	1.125	1.260	0.688	-0.042
D.VER	1.083	1.938	1.313	1.583	0.573	0.063
D.HOR	1.354	2.271	1.906	1.823	0.833	-0.427

Wilcoxon signed-rank test with Bonferroni correction revealed a statistically significant difference in *RowDiff* in the middle part of the text between C.BUT and D.VER ($p = .015$). However, the absolute distance to the correct row (*RowDiff* = 0) is similar between C.BUT (*RowDiff* = -5.188) and D.VER (*RowDiff* = +4.396). We also looked at the absolute row difference (absolute number from correct row) and performed an Aligned Rank Transform (ART) ANOVA. The result indicates that text panel variants do not have a significant effect on absolute row difference and that there is no significant interaction between the variants and the text parts. The result also suggests that there is a significant effect ($p < .0001$) from position of the information (*beginning*, *middle*, *end*) on the absolute row difference. The ART-C post hoc pairwise comparisons with Holm's sequential Bonferroni correction indicate that (C.BAR, *beginning*) in contrast to (C.BUT, *end*) are significantly different ($p = .048$). All other pairwise comparisons are not significantly different.

5.3 Simulator Sickness Questionnaire (SSQ)

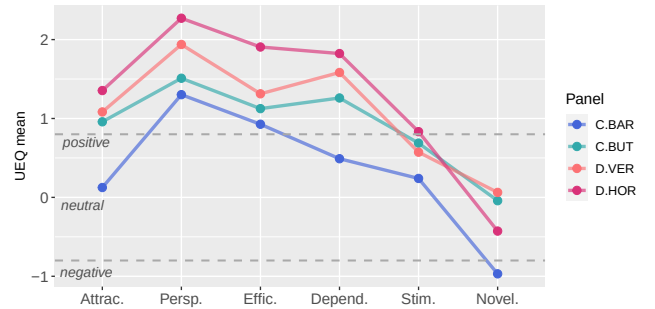
The results in Figure 4 show an increase in *Mean* scores on all SSQ scales except for *Nausea*. Since normal distribution for the data could not be assumed (Shapiro-Wilk test), we performed Friedman tests for all three SSQ categories and for the *Total Simulator Sickness TS* score. We did not find a significant difference in SSQ scores between the trials (including baseline). While reading of long texts in VR increased symptoms, especially on the *Oculomotor Disturbance* scale ($M = 12.95$ at baseline, 19.90 after trial 4), it did not lead to a significant worsening of these symptoms.

**Figure 4: Mean SSQ scores after each trial for each SSQ category, including the total score**

5.4 User Experience Questionnaire (UEQ)

Overall, the UEQ results are positive. As seen in Table 2 the scales *Attractiveness*, *Perspicuity*, *Efficiency*, and *Dependability* mainly received positive ratings (with > 0.8 considered as positive, -0.8 to

0.8 neutral and < -0.8 negative in UEQ rating). The hedonic subscales *Stimulation* and *Novelty* were mainly rated neutrally, with D.HOR receiving a positive rating for *Stimulation* (0.833) and C.BAR receiving a negative rating for *Novelty* (-0.969). In comparison to the UEQ benchmark data set, only D.HOR received an excellent rating on the pragmatic scales, while C.BUT and D.VER achieved a good rating. All variants got a low rating on the *Novelty* scale compared to the benchmark. This rating is illustrated accordingly in Figure 5 which visualizes the placement of the variants with respect to each other for better comparison. C.BAR received the lowest ratings on all subscales, while D.HOR received the highest ratings on all subscales except for *Novelty*. D.VER is in second place and C.BUT places third in the UEQ comparison. Although C.BUT has a slightly higher rating on the *Stimulation* scale compared to D.VER, it scored lower on all the other scales.

**Figure 5: Comparison of the Mean ratings between the panel variants on all UEQ scales**

The *Means* for *Attractiveness* and *Efficiency* are normally distributed. We conducted a one-way repeated measurements ANOVA and found significant differences between the text panel variants for *Attractiveness* ($p = .002$) and *Efficiency* ($p = .006$). The pairwise comparison post-hoc *t*-test with Bonferroni correction found statistically significant differences between the following text panel variants:

- D.HOR has a significantly higher *Attractiveness* score than C.BAR ($p = .007$)
- D.HOR has a significantly higher *Efficiency* score than C.BAR ($p = .003$)
- D.HOR has a significantly higher *Efficiency* score than C.BUT ($p = .035$)

For the *Means* of the remaining UEQ scales, a normal distribution could not be assumed based on the Shapiro-Wilk test. A Friedman tests indicated significant differences for *Perspicuity* ($p = .019$),

Dependability ($p = .02$), and *Novelty* ($p = .04$). Wilcoxon post-hoc tests with Bonferroni correction revealed the following statistically significant differences:

- **D.HOR** has a significantly higher *Perspicuity* score than **C.BAR** ($p = .007$)
- **D.HOR** has a significantly higher *Dependability* score than **C.BAR** ($p = .029$)
- **C.BUT** has a significantly higher *Novelty* score than **C.BAR** ($p = .017$)

5.5 Qualitative Results

Subjective Ranking. Participants were asked to rank the text panel variants from highest (1) to lowest (4). As illustrated in Figure 6, there was no clear preference for one variant. **C.BUT**, **D.VER**, and **D.HOR** got about a third of the votes each for rank 1, while only four participants voted for **C.BAR**. However, for the least preferred variant (rank 4), 16 participants (66.7 %) chose **C.BAR** and 6 (25 %) chose **D.VER**.

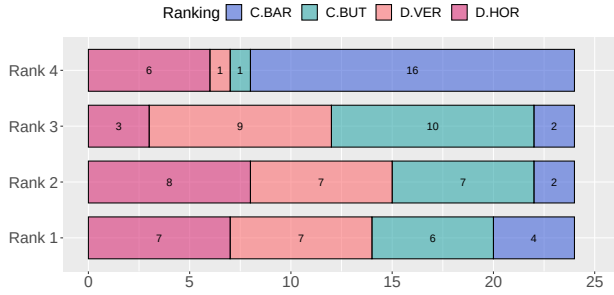


Figure 6: Subjective ranking with number of votes per text panel variant for each rank

Qualitative User Feedback. We used a structured, inductive qualitative content analysis based on Mayring [14]. Using inductive reasoning, we repeatedly analysed and compared the qualitative feedback to identify reappearing themes/topics in the statements of the participants. Finally, we consolidated the these themes into the following categories: *placement*, *usability*, *view*, *orientation*, and *animation*. We systematically assigned the statements on each text panel variant to these categories. Corresponding statements within a category were paraphrased and counted. An overview of positive and negative statements with the highest occurrences ($N > 3$) for each text panel variant is listed in Table 3. **C.BAR** received negative statements with the highest overall occurrence and no positive statement with $N > 3$. Both negative statements are related to the bad usability of the scrollbar. More than half of the participants ($N = 14$) expressed that the scrollbar is too imprecise/sensitive, and 9 stated that the jumping text leads to a loss of orientation when scrolling. The other three variants received positive overall feedback in terms of usability. But again, the usability of the scrollbar in **C.BUT** was criticised. Some participants stated that the fast scrolling with the up and down buttons resulted in a loss of orientation, while others enjoyed the speed and smoothness of the scrolling. Both discrete variants received positive feedback for

Table 3: Qualitative content analysis – Positive and negative statements with the highest occurrences

Panel	Overview of Statements	N
C.BAR	- scrollbar imprecise/too sensitive, difficult to use/exhausting	14
	- jumping scrollbar: loss of orientation/impaired reading flow/searching for text passages	9
C.BUT	+ simple, intuitive	5
	+ text comfortably readable at eye level	4
	+ smooth, pleasant speed of scrolling with up/down buttons	4
	- scrollbar too small and too sensitive	4
D.VER	- loss of orientation due to fast button scrolling	4
	+ easy to use, intuitive, straightforward	6
	+ control elements on right side easy to access	4
	+ clear structure through pagination	4
D.HOR	- loss of orientation when changing pages; text between two pages not visible at the same time	4
	- vertical pagination animation is not-intuitive, irritating	4
	+ simple, easy to use, straightforward	8
	- position of control elements at bottom of panel is impractical/too far away	7
General	+ clear structure through pagination	5
	+ book metaphor is good	4
General	• additional control of texts with thumbstick, e.g. for scrolling and pagination	5
	• adjusting scrolling speed with thumbstick or with trigger button pressure	5

the clear structure provided by the pagination. But participants reported loss of orientation as well as unintuitive and irritating animation for the vertical pagination in **D.VER**. While some statements appreciated the placement of the UI controls for **D.VER** for being easy to access, the position of the UI controls for **D.HOR** at the bottom of the text panel was criticised for being impractical and too far away. Nevertheless, four participants stated that they liked the book metaphor of **D.HOR**.

As general feedback, participants requested additional controls for scrolling and pagination, e.g. with the thumbstick of the VR controller. Moreover, they suggested adjusting the scrolling speed with the thumbstick or the trigger button, so that the reading flow is not disrupted by looking at and interacting with GUI controls in VR.

6 DISCUSSION AND CONCLUSION

Our results do not support (H1) since we did not find a difference in *Recall* between continuous and discrete text panel variants. However, this finding is consistent with Chaparro and Baker's [1] findings for reading on desktop computers. It should therefore be further investigated if other existing research findings for reading on desktop also apply to reading in VR. Further insights in this

area could help with transferring and adapting already established desktop UI patterns for VR. Our findings also do not support (H2). Although there was a significant difference in *Spatial Recall* between **C.BUT** (RowDiff = -5.188) and **D.VER** (RowDiff = +4.396) in the middle part of the text, the absolute row difference was not significant, so no variant performed more accurate or better at this task. Although there is no statistical significance, it is interesting to see in Figure 3 how participants tend to underestimate the location of information at the end of the text. This observation and possible causes could be investigated in future studies on reading in VR. Moreover, our results suggest that some serial position effects might apply for reading in VR. As a limitation, it has to be noted that despite the careful selection of our text parameters and questions, we cannot rule out the possibility that the different texts or questions may have influenced the results.

Based on our results, we can accept (H3). Both discrete variants received higher ratings for *Attractiveness* and for the three pragmatic UEQ scales, with **D.HOR** having significantly higher ratings than **C.BAR** on all these scales. All variants received a below average or bad rating in the UEQ benchmark in terms of the hedonic qualities, *Stimulation* and *Novelty*. This was expected since the designs were based on familiar desktop UI patterns. Nonetheless, all variants except for **C.BAR** received good or even excellent ratings for the pragmatic quality compared to the benchmark data set. **C.BAR** received the overall lowest scores of all variants. Our additional qualitative feedback supports the UEQ findings. The subjective ranking showed no clear preference for one variant, but the majority ranked **C.BAR** in the last place (66.7 %). The Qualitative Content Analysis of the participants' feedback gave us valuable insights into the cause of the bad rating for **C.BAR**. The issue seems to be the poor usability of the scrollbar in VR. Several participants stated that the discrete variants were easy to use, intuitive, and had a clear structure due to the pagination. Based on our findings, we recommend using the pagination pattern instead of a scrollbar for displaying and interacting with long texts in VR. Participants named different aspects they liked and disliked for each of the text panel variants. They also included feature suggestions, such as using the VR controller's thumbstick to control the text flow and speed. The user feedback suggests that it is important to offer alternative interaction methods, including non GUI ones, for reading long texts in VR. In future work, improved scrollbars and additional alternative UI control methods for the text panels could be designed and developed to address issues with the current text panel variants. Accordingly, the performance and UX should be re-evaluated for discrete and continuous text representations in VR.

Continuous and discrete versions each seem to have their advantages and disadvantages. Depending on the use case or individual preference, a certain panel variant may be favoured by users. So far, we have looked at performance and UX, with participants mainly coming from our computer science faculty. Follow-up evaluations with researchers from different fields in the humanities could provide more specific results and insights on reading long texts in VR for the specific use cases in our project.

To summarize, while we did not find significantly better reading performance or a clear user preference between the variants, we observed that desktop UI patterns can also work for reading long texts in VR. As for the pragmatic aspects, **D.HOR** with its horizontal

pagination received the highest UX ratings. Our findings indicate that patterns and guidelines for reading long texts in VR need to be further investigated. Adapted UI patterns and interaction design should be improved and user preferences depending on the use case should be further examined. For future evaluations, we plan to collect additional quantitative data, such as eye-tracking data, to gain insights on reading behaviour in VR. This would also allow us to distinguish between time actually spent reading and time spent viewing and interacting with the user interface. Our work also demonstrates the importance of following a user-centred design approach and the collection of qualitative user feedback for research on UI patterns in VR. The additional information helped us to better understand user preferences and pain points. We believe that these findings will facilitate our future development and research on suitable UI patterns for reading long texts in VR.

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